Continuous Keyframe-Free Holographic Compression for Real-Time Audio-Video Streaming over Ultra-Low Bandwidth 2G Networks

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Abstract

This paper presents a real-time audio-video streaming system capable of delivering stable 640×480 resolution at 25 fps with synchronized audio over genuine 2G GSM networks (9–20 kbps, high packet loss, high latency).

The proposed method departs from conventional predictive codecs by eliminating keyframes entirely and introducing a continuous holographic compression model based on high-dimensional geometric embedding, adaptive multi-view projection, and sparse reconstruction.

The system exhibits intrinsic resilience to packet loss and bandwidth degradation, achieving graceful quality scaling instead of freezing or collapse. Experimental observations further indicate that Forward Error Correction (FEC) is unnecessary and may reduce effective quality in this regime.

Keywords: holographic compression, keyframe-free streaming, sparse representation, low-bandwidth video, 2G networks, adaptive streaming

1 Introduction

Video streaming over ultra-low bandwidth and unstable networks remains a critical challenge. Conventional codecs rely on keyframes and inter-frame dependencies, which introduce fragility under packet loss and bandwidth fluctuations, often leading to freezing or complete stream collapse.

This work proposes a fundamentally different paradigm: a continuous, keyframe-free representation that degrades gracefully instead of failing abruptly.

2 System Pipeline

The pipeline is defined as:

1. Frame acquisition via OpenCV / FFmpeg
2. Partition into blocks B_i
3. High-dimensional embedding $\mathbf{p}_i = \phi(B_i)$
4. Temporal difference $\mathbf{v}_i$
5. Adaptive multi-view projection
6. Sparse coding (OMP over DCT dictionary)

7. Quantization
8. Entropy compression (Zstandard)
9. Packetization (UDP)
10. Streaming via MediaMTX (RTSP/WebRTC relay)
11. Reconstruction and rendering

3 Mathematical Formulation of Holographic Compression

Let a video frame be represented as:

$$\mathbf{F}_t \in \mathbb{R}^{H \times W \times C}$$

It is partitioned into patches:

$$\mathbf{F}_t \to \{B_i\}_{i=1}^{N}$$

Each patch is mapped into a high-dimensional latent space:

$$\phi : \mathbb{R}^n \to \mathbb{R}^D$$

$$\mathbf{p}_i = \phi(B_i)$$

where $D \gg n$, enabling redundant distributed encoding.

3.1 Holographic Encoding Principle

The representation is defined such that information is not localized but distributed:

$$\mathcal{H}(\mathbf{p}_i) = \{\Pi_k \mathbf{p}_i\}_{k=1}^{K}$$

where $\Pi_k \in \mathbb{R}^{d \times D}$ are projection operators.

Each projection contains partial information about the full signal.

3.2 Reconstruction from Partial Observations

Given a subset $\mathcal{K} \subseteq \{1, \ldots, K\}$:

$$\hat{\mathbf{p}}_i = \arg\min_{\mathbf{p}} \sum_{k \in \mathcal{K}} \|\Pi_k \mathbf{p} - \mathbf{y}_k\|_2^2$$

This ensures robustness to packet loss, as reconstruction remains feasible even with missing projections.

3.3 Connection to Compressed Sensing

Assuming sparsity in a basis D:

$$\mathbf{p}_i = D\mathbf{x}_i, \quad \|\mathbf{x}_i\|_0 \ll D$$

We observe:

$$\mathbf{y}_k = \Pi_k D \mathbf{x}_i$$

Reconstruction becomes:

$$\min_{\mathbf{x}} \|\mathbf{x}\|_0 \quad \text{s.t.} \quad \mathbf{y} = A\mathbf{x}, \quad A = \Pi D$$

This aligns the system with compressed sensing theory, ensuring recoverability under partial measurements.

4 Temporal Adaptation

Frame variation:

$$\Delta_t = \|\mathbf{F}_t - \mathbf{F}_{t-1}\|_2^2$$

$$\text{mode} = \begin{cases} \text{delta} & \text{if } \Delta_t < xdel \\ \text{full} & \text{if } \Delta_t \geq xdel \end{cases}$$

5 Patch-Level Adaptation

$$\delta_j = \|\mathbf{P}_{j,t} - \mathbf{P}_{j,t-1}\|_2^2$$

$$\text{mode}_j = \begin{cases} \text{skip} & \text{if } \delta_j < ydel \\ \text{update} & \text{if } \delta_j \geq ydel \end{cases}$$

6 Holographic Temporal Dynamics

Instead of classical prediction:

$$\mathbf{v}_i = \mathbf{p}_{i,t} - \mathbf{p}_{i,t-1}$$

Projected dynamics:

$$\mathbf{v}_i^{(k)} = \Pi_k \mathbf{v}_i$$

Reconstruction:

$$\hat{\mathbf{v}}_i = \arg\min_{\mathbf{v}} \sum_k \|\Pi_k \mathbf{v} - \mathbf{v}_i^{(k)}\|_2^2$$

Final update:

$$\hat{\mathbf{p}}_{i,t} = \mathbf{p}_{i,t-1} + \hat{\mathbf{v}}_i$$

7 Sparse Coding

$$D = D_{1D} \otimes D_{1D}$$

$$\min_{\mathbf{x}} \|\mathbf{y} - D\mathbf{x}\|_2^2 \quad \text{s.t.} \quad \|\mathbf{x}\|_0 \leq K$$

8 Adaptive Bitrate Behavior

$$R(t) = \sum_j R_j(t), \quad R(t) \leq B(t)$$

Adaptation:

- sparsity $\downarrow$
- quantization $\downarrow$
- skip rate $\uparrow$

9 On the Ineffectiveness of FEC

$$R_{\text{total}} = R_{\text{data}} + R_{\text{fec}}$$

$$R_{\text{data}} = B(t) - R_{\text{fec}}$$

Increasing FEC reduces signal quality. Due to the absence of keyframes and long dependencies, packet loss only produces local degradation.

10 System Properties

- Keyframe-free
- Continuous degradation
- Immediate recovery
- Intrinsic robustness

11 Applications in Constrained Environments

11.1 Video Surveillance

Continuous monitoring in low-coverage areas becomes feasible without freezing.

11.2 Emergency Drone Streaming

Provides stable visual feedback under degraded network conditions.

11.3 Telemedicine

Enables continuous doctor-patient communication over legacy networks.

12 Limitations and Future Work

The current system prioritizes robustness over visual fidelity. Future work includes:

- objective quality evaluation (PSNR, SSIM)
- GPU acceleration
- optimization of projection selection
- large-scale benchmarking vs standard codecs

13 Conclusion

A continuous, keyframe-free holographic compression system has been presented for ultra-low bandwidth streaming. The approach replaces fragile predictive structures with a distributed representation that inherently absorbs packet loss and adapts to network conditions, enabling stable real-time communication where traditional systems fail.